19. Letter Sequences & Alphabet Code (Set B)

Year 4 · **Supplementary Worksheet** — extra practice & worked breakdowns

Worked Examples

Example 1 — Letter sequence:

What comes next in the sequence: D, E, F, G, ?

Answer: Each letter moves forward 1 place (+1 each time): D, E, F, G. G + 1 = H. Answer: **H**.

Example 2 — Position code:

If A=1, B=2, C=3 and so on, what number stands for the letter J?

Answer: J is the 10th letter of the alphabet. Answer: **10**.

Example 3 — Shift code with a worked pair:

A puzzle gives you this pair as a worked example of a code: CUP → EWR. Work out the shift being used, then use it to encode the word 'DOG'.

Answer: C→E is +2, U→W is +2, P→R is +2, so the shift is +2. Applying +2 to DOG: D→F, O→Q, G→I. Answer: **+2, so DOG becomes FQI**.

Example 4 — Multiple-choice:

What comes next in the sequence: F, H, J, L, ? A) M B) N C) O D) P

Answer: The gap is +2 throughout, so L + 2 = N. Answer: **B) N**.

Example 5 — Select-all:

Select ALL that apply: which of these letter pairs are exactly 3 letters apart in the alphabet (e.g. B and E)? A) C and F B) G and J C) K and N D) P and R E) S and V

Answer: C–F is 3 apart (A), G–J is 3 apart (B), K–N is 3 apart (C), and S–V is 3 apart (E). P–R is only 2 apart, so D is not correct. Answer: **A, B, C and E**.

Foundation

15 questions

1. What comes next in the sequence: D, E, F, G, ?

2. What comes next in the sequence: K, J, I, H, ?

3. If A=1, B=2, C=3..., what number stands for the letter J? A) 9 B) 10 C) 11 D) 8

4. What comes next in the sequence: F, H, J, L, ?

5. What comes next in the sequence: W, U, S, Q, ?

6. In a code where A=1, B=2, C=3..., which letter stands for the number 9? A) H B) I C) J D) G

7. Fill in the missing letter: B, _, F, H, J (skipping one letter each time)

8. If every letter is shifted forward by 1 place (A→B, B→C, C→D...), what does E become? A) D B) F C) G D) C

9. What comes next in the sequence: G, H, I, J, ?

10. Select ALL that apply: which of these letter pairs are exactly 2 letters apart in the alphabet (e.g. B and D)? A) E and G B) H and K C) M and O D) P and S E) V and X

11. What comes next in the sequence: O, N, M, L, ?

12. If A=1, B=2, C=3..., what number stands for the letter P? A) 15 B) 16 C) 17 D) 14

13. What comes next in the sequence: C, E, G, I, ?

14. If every letter is shifted back by 1 place (B→A, C→B, D→C...), what does J become? A) I B) K C) H D) L

15. What comes next in the sequence: R, Q, P, O, ?

Standard

20 questions

1. What comes next in the sequence: D, G, J, M, ?

2. What comes next in the sequence: V, S, P, M, ?

3. What comes next in the sequence: F, J, N, R, ? A) U B) V C) W D) X

4. What comes next in the sequence: X, U, R, O, ?

5. In a position code (A=1...Z=26), what is the sum of the position numbers of the letters in the word LOG?

6. A code shifts every letter forward by 4 places (A→E, B→F...). What does the word 'DEN' become? A) HIR B) HIS C) GIR D) HJR

7. What comes next in the sequence: Z, V, R, N, ?

8. In a code, A=1, B=2, C=3... Two numbers are added together and converted back to a letter: if the numbers are 5 and 9, which letter results? A) M B) N C) O D) L

9. Select ALL that apply: which sequences below correctly continue with a GAP of 4 between each letter? A) B, F, J, N, R B) C, G, K, O, S C) D, H, K, O, S D) E, I, M, Q, U E) A, E, H, L, P

10. What comes next in the sequence: W, S, O, K, ?

11. A code shifts every letter BACK by 4 places (E→A, F→B...). What does the word 'KSV' decode to? A) GOR B) GOQ C) HOR D) GPR

12. What comes next in the sequence: A, D, G, J, ?

13. What comes next in this sequence, where the gap increases by 1 each time (+3, +4, +5...)? B, E, I, N, ?

14. What comes next in this sequence, which alternates two interleaved patterns? B, Y, C, X, D, ? A) W B) V C) U D) X

15. Select ALL that apply: in a +6 shift code (A→G, B→H, C→I...), which of these letter pairs are correctly matched? A) A→G B) D→J C) H→N D) M→S E) Y→D (wrapping to the start of the alphabet)

16. What comes next in the sequence: U, Q, M, I, ?

17. In a position code (A=1...Z=26), what is the sum of the position numbers of the letters in the word WEB?

18. What comes next in the sequence: J, N, R, V, ?

19. What comes next in this sequence, where the gap increases by 1 each time (+2, +3, +4...)? C, E, H, L, ? A) P B) Q C) O D) N

20. For the sequence H, L, P, T, which of these are true? A) The gap between each letter is 4. B) The next letter would be X. C) The sequence goes backwards through the alphabet. D) The gap between L and P is 4. E) The first letter is H.

Hard

5 questions

1. What comes next in the sequence: C, H, M, R, ?

2. What comes next in this sequence (wrapping round after Z): V, A, F, K, ? A) O B) P C) Q D) N

3. Select ALL that apply: in a −5 shift code (F→A, G→B, H→C…), which of these letter pairs are correctly matched (wrapping round the start of the alphabet where needed)? A) J→E B) M→H C) C→X D) D→Y E) B→V

4. A puzzle gives you this pair as a worked example of a code: LION → OLRQ. Work out the shift being used, then use it to encode the word ‘BEAR’.

5. A puzzle gives you this pair as a worked example of a code: FROG → COLD. Work out the shift being used, then decode the word ‘ZLVK’ using the same shift.

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